

Communications Network



Multiplexing is a method of sending 2 or more signals simultaneously over a single circuit. Multiplexing allows 2 or more electronic modules (nodes) to communicate simultaneously over a twisted wire pair [data (+) and data (-)] network. The information or messages that can be communicated on these wires consists of commands, status or data. The advantage of using multiplexing is to reduce the weight of the vehicle by reducing the number of redundant components and electrical wiring.

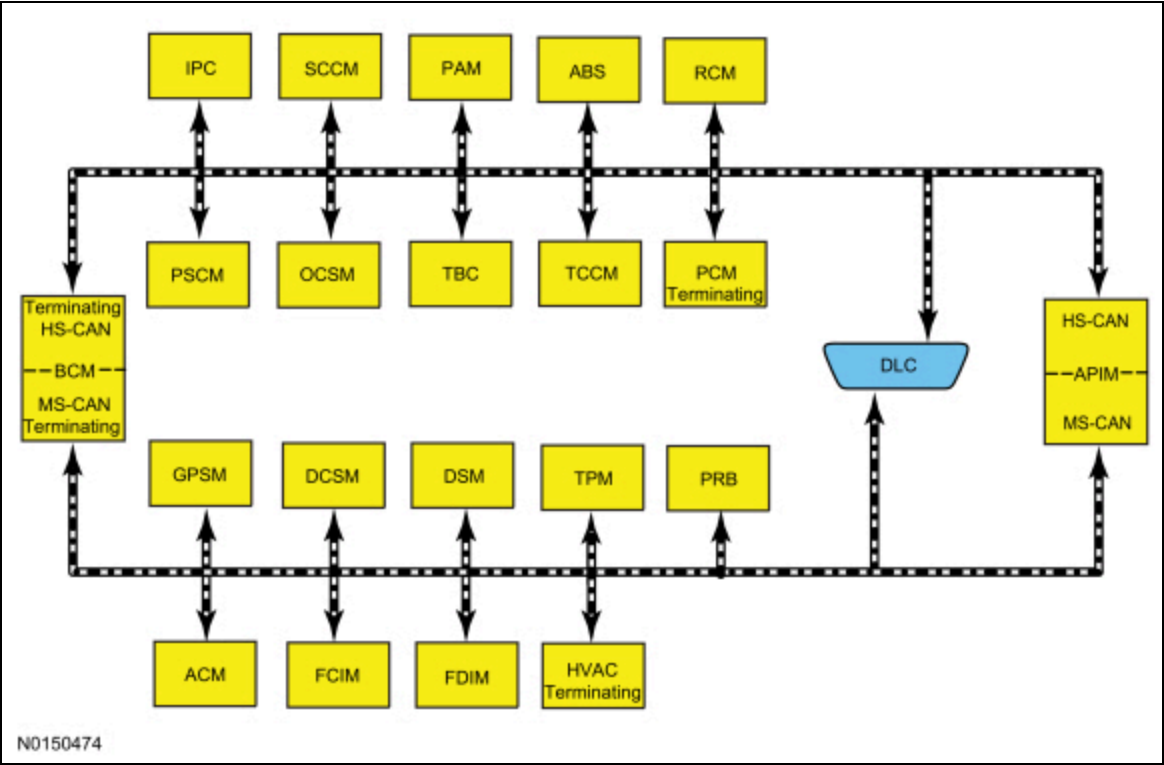
The vehicle has 3 module communication networks which are connected to the Data Link Connector (DLC) , located under the driver side instrument panel:

- High Speed Controller Area Network (HS-CAN)
- Medium Speed Controller Area Network (MS-CAN)
- Infotainment Controller Area Network (I-CAN) (4.2-inch and 8-inch touchscreen only)

NOTE: Vehicles with the optional 4.2-inch screen and 8-inch touchscreen audio units have an Infotainment Controller Area Network (I-CAN) that is connected to the Data Link Connector (DLC) , but modules do not communicate directly with the diagnostic scan tool. The Instrument Panel Cluster (IPC) operates as the gateway for modules on the I-CAN to communicate with the diagnostic scan tool over the HS-CAN .

NOTE: The TPM module is also known as the Radio Transceiver Module (RTM).

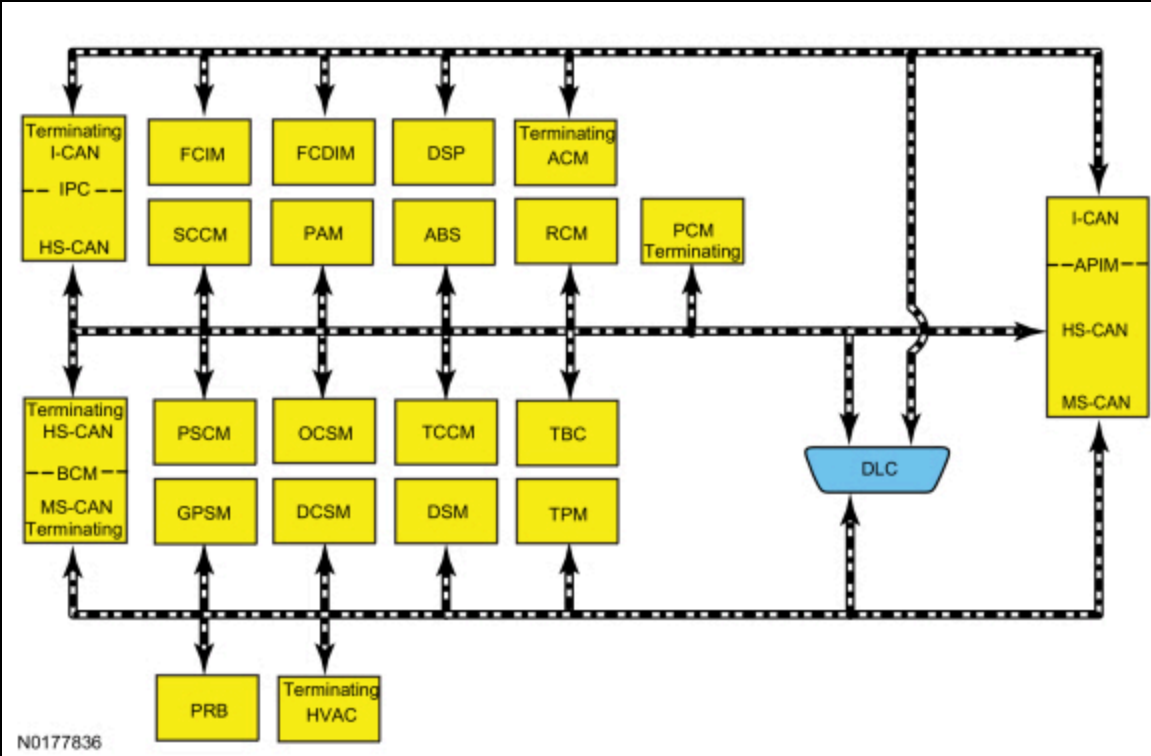
Network Topology - Vehicles with AM/FM/Single CD



Module Name	Network Type	Termination Module
ABS module	HS-CAN	No

Module Name	Network Type	Termination Module
Accessory Protocol Interface Module (APIM) (if equipped)	HS-CAN	No
	MS-CAN	No
Audio Front Control Module (ACM) (if equipped)	MS-CAN	No
Body Control Module (BCM) (gateway module)	MS-CAN	Yes
	HS-CAN	Yes
Driver Seat Module (DSM) (if equipped)	MS-CAN	No
Dual Climate Controlled Seat Module (DCSM) (if equipped)	MS-CAN	No
Front Controls Interface Module (FCIM) (if equipped)	MS-CAN	No
Front Display Interface Module (FDIM)	MS-CAN	No
Global Positioning System Module (GPSM) (if equipped)	MS-CAN	No
HVAC module	MS-CAN	Yes
Instrument Panel Cluster (IPC)	HS-CAN	No
Occupant Classification System Module (OCSM)	HS-CAN	No
Parking Aid Module (PAM) (if equipped)	HS-CAN	No
Power Steering Control Module (PSCM) (if equipped)	HS-CAN	No
PCM	HS-CAN	Yes
Power Running Board (PRB) module (if equipped)	MS-CAN	No
Restraints Control Module (RCM)	HS-CAN	No
Steering Column Control Module (SCCM)	HS-CAN	No
Tire Pressure Monitor (TPM) module (Radio Transceiver Module [RTM]) (if equipped)	MS-CAN	No
Trailer Brake Control (TBC) module (if equipped)	HS-CAN	No
Transfer Case Control Module (TCCM) (if equipped)	HS-CAN	No

Network Topology - Vehicles with 4.2-Inch Screen and 8-Inch Touchscreen



Module Name	Network Type	Termination Module
ABS module	HS-CAN	No
Accessory Protocol Interface Module (APIM) (if equipped)	HS-CAN	No
	MS-CAN	No
	I-CAN	No
Audio Front Control Module (ACM) (if equipped)	I-CAN	Yes
Audio Digital Signal Processing (DSP) module (if equipped)	I-CAN	No
Body Control Module (BCM) (gateway module)	MS-CAN	Yes
	HS-CAN	Yes
Driver Seat Module (DSM) (if equipped)	MS-CAN	No
Dual Climate Controlled Seat Module (DCSM) (if equipped)	MS-CAN	No
Front Controls Interface Module (FCIM) (if equipped)	I-CAN	No
Front Control/Display Interface Module (FCDIM) (if equipped)	I-CAN	No
Global Positioning System Module (GPSM) (if equipped)	MS-CAN	No
HVAC module	MS-CAN	Yes
Instrument Panel Cluster (IPC)	HS-CAN	No
	I-CAN	Yes
Occupant Classification System Module (OCSM)	HS-CAN	No
Parking Aid Module (PAM) (if equipped)	HS-CAN	No
Power Steering Control Module (PSCM) (if equipped)	HS-CAN	No
PCM	HS-CAN	Yes
Power Running Board (PRB) module (if equipped)	MS-CAN	No
Restraints Control Module (RCM)	HS-CAN	No
Steering Column Control Module (SCCM)	HS-CAN	No

Module Name	Network Type	Termination Module
Tire Pressure Monitor (TPM) module (Radio Transceiver Module [RTM]) (if equipped)	MS-CAN	No
Trailer Brake Control (TBC) module (if equipped)	HS-CAN	No
Transfer Case Control Module (TCCM) (if equipped)	HS-CAN	No

Network Termination And Gateway Function

The Controller Area Network (CAN) uses network termination to improve communication reliability. Termination modules are located at both ends of the network. As network messages are broadcast in the form of voltage signals, the network voltage signals are stabilized by the termination resistors.

Each termination module has an internal 120 ohm resistor that bridges across the positive and negative bus connection. With two 120 ohm resistors located in a parallel circuit configuration, the total network impedance, or total resistance, is 60 ohms.

Network termination improves bus message reliability by stabilizing bus voltage and eliminating electrical interference.

The BCM is the gateway module, translating messages on the HS-CAN to the MS-CAN and vice versa, allowing a message to be distributed throughout both networks.

Vehicles equipped with the 4.2-inch screen or 8-inch touchscreen audio units use an I-CAN network for these modules. The IPC is the gateway module, translating messages on the HS-CAN to the modules on the I-CAN , and vice versa. This allows messages to be distributed throughout both networks, if required.

The BCM and IPC are the only modules on this vehicle that have this ability.